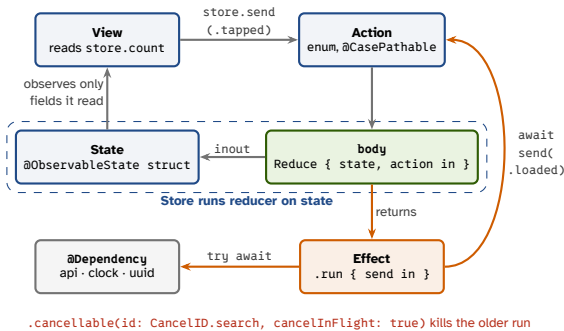
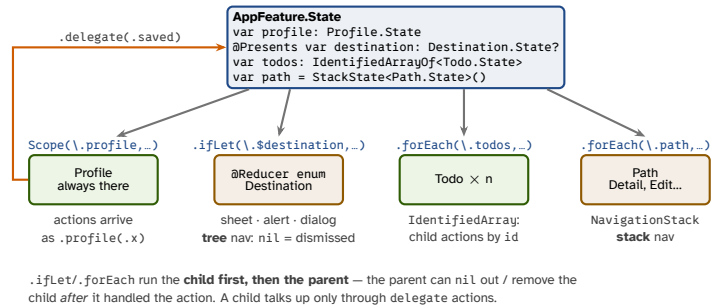


In one line: Point-Free's library for **unidirectional** Swift apps: a feature is a **@Reducer** — value-type **State**, an **Action** enum, a **body** that mutates state and returns **Effects**; a **Store** runs it; features **compose** by embedding child state/actions; the outside world comes in through **@Dependency**; **TestStore** proves every step. Current **1.26.2** (2026-08-28).

One feature: the loop the Store runs



Composition: children live *inside* parent state



How it works

- @Reducer:** nested State (@ObservableState = Observation for structs), Action (made @CasePathable → \.loaded), body: some ReducerOf<Self>.
- Store:** Store(initialState: F.State()) { F() }; view: let store: StoreOf<F> · store.x · store.send(.a) · store.scope(\.kid, action: \.kid).
- Effects:** .none · send(.a) · run { send in } · merge · cancellable(id: cancelInFlight: true) · cancel(id:).
- Dependencies:** @DependencyClient struct of closures + DependencyKey (liveValue, testValue, previewValue); read @Dependency(\.api); override with withDependencies. An unset dependency called in a test **fails** it.
- Tree nav:** @Presents var destination + PresentationAction<...> + .ifLet(\.\$destination, action: \.destination); view .sheet(item: \$store.scope(...)); child closes with @Dependency(\.dismiss).
- Stack nav:** StackState<Path.State> + StackActionOf<Path> + .forEach(\.path, action: \.path); NavigationStack(path: \$store.scope(...)). Deep link = construct the state.
- Bindings:** @Bindable var store + BindableAction / BindingReducer() → \$store.name; or \$store.name.sending(\.nameChanged).
- @Shared** (swift-sharing): .appStorage("k"), .fileStorage(url), .inMemory("k"); write via \$x.withLock { \$0 += 1 } (+= is not atomic).
- 1.25–1.26:** trait ComposableArchitecture2Deprecations flags 2.0-bound APIs (BindingViewStore, Effect.map); 1.26: Scope(\.kid, action:).

Worth it?

Pays: big team wanting one shape; navigation/deep links as state; many cancellable effects; exhaustive tests. **Costs:** learning curve, boilerplate, **compile time** (macros + swift-syntax, deep generic reducers), lock-in + migrations (2.0), every high-frequency event (scroll, drag) walks the reducer tree.

Remember

State in the struct, **effects** in `.run`, the **world** in `@Dependency`; **scope** down, **delegate** up.

Example — debounced search + its test

```
@Reducer struct Search {
  @ObservableState struct State: Equatable {
    var query = ""; var results: [Item] = []
  }
  enum Action { case queryChanged(String), loaded([Item]) }
  enum CancelID { case search }
  @Dependency(\.api) var api
  @Dependency(\.continuousClock) var clock
  var body: some ReducerOf<Self> {
    Reduce { state, action in
      switch action {
      case let .queryChanged(q):
        state.query = q
        return .run { send in
          try await clock.sleep(for: .milliseconds(300))
          await send(.loaded(try await api.search(q)))
        }, cancellable(id: CancelID.search, cancelInFlight: true)
      case let .loaded(items):
        state.results = items; return .none
      }
    }
  }
}

// test - exhaustive by default: every change asserted
let clock = TestClock()
let store = TestStore(initialState: Search.State()) { Search() }
withDependencies: { $0.continuousClock = clock
  $0.api.search = { _ in [.mock] } }
await store.send(.queryChanged("sw")) { $0.query = "sw" }
await clock.advance(by: .milliseconds(300))
await store.receive(\.loaded) { $0.results = [.mock] }
```

Unasserted change, unreceived action or a still-running effect ⇒ failure.
store.exhaustivity = .off = assert only what matters (big integration flows).

Traps

- ! Task {} / a singleton inside Reduce — untestable, uncancellable. Return `.run`; reach the world via `@Dependency`.
- ! Search without `cancelInFlight` — a slow old response overwrites a newer one.
- ! Sharing logic by *sending actions* — each hop re-runs the reducer tree; call a helper and return one effect.
- ! Parent matching a child's *internal* actions — listen to `.delegate` only.